

DAX IN POWER BI

What is Dax

Ans: - DAX (Data Analysis Expressions) is a formula language used in Power BI, Power Pivot (Excel), and SQL Server Analysis Services (SSAS) to perform data modeling and advanced calculations.

It is designed for:

- Creating custom calculations in Power BI reports.
- Building calculated columns, measures, and tables.
- Performing time intelligence, filtering, and aggregation operations.

Why is DAX Important?

Ans: - DAX is important for custom calculations as it helps create new values like total sales or profit.

DAX is important for time-based analysis because it allows comparing data like this year's vs. last year's sales.

DAX is important for dynamic reports as it updates results automatically based on user selections.

DAX is important for better filtering because it helps control what data is shown using special functions.

DAX Syntax

Ans: - DAX formulas follow a structured syntax similar to Excel but are designed for relational data models.

Basic Syntax Structure: -

Measure_Name = Function (Table [Column])

Example: -

Total_Sales = SUM(SalesData[Sales])

(This formula calculates the total of all numbers in the Sales column from the SalesData table.)

DAX Calculation Types

Ans: - DAX formulas are used to create: -

- **Calculated Columns** (Row-level calculations)
- **Measures** (Aggregation calculations)
- **Calculated Tables** (Custom table creation using DAX).

Calculated Columns (Row-level calculations)

Ans: - A calculated column adds a new column to a table, performing row-by-row calculations.

When to use?: Use calculated columns when you need a new data field that depends on other column values in the same row.

Example:

`Total_Price = SalesData[Quantity] * SalesData[Price]`

Measures (Aggregation calculations)

Ans: - A measure performs dynamic aggregations like sums, averages, or counts based on filters in visuals.

When to use?: Use measures when you want calculations that change based on report filters.

Example:-

`Total_Sales = SUM(SalesData[Sales])`

(This measure dynamically calculates total sales, updating automatically based on applied filters.)

Calculated Tables (Custom table creation using DAX)

Ans: - A calculated table generates a new table based on an existing dataset using DAX.

When to use?: Use calculated tables to create filtered, summarized, or new structured datasets.

Example: -

`High_Sales_Regions = FILTER(SalesData, SalesData[Sales] > 50000)`

(This creates a new table containing only regions where total sales exceed 50,000.)

DAX Functions & Examples

Aggregation Functions: -

Aggregation functions are used to summarize and calculate values from a column in a dataset. Below are the key aggregation functions with examples.

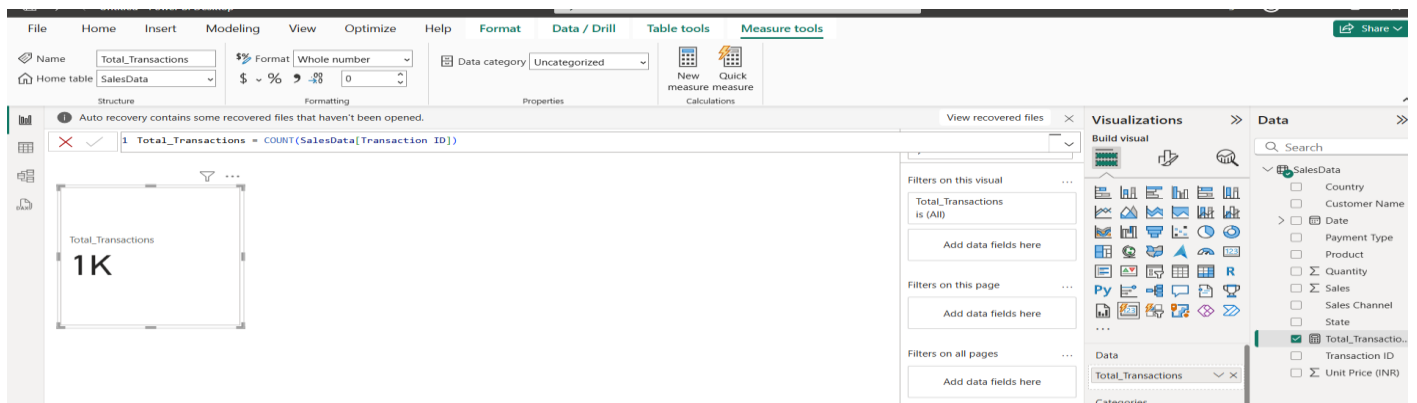
1) COUNT Functions.

These functions count the number of rows or distinct values in a column.

COUNT: - Counts the number of rows in a column

Example: - counts the total number of transactions in the SalesData table.

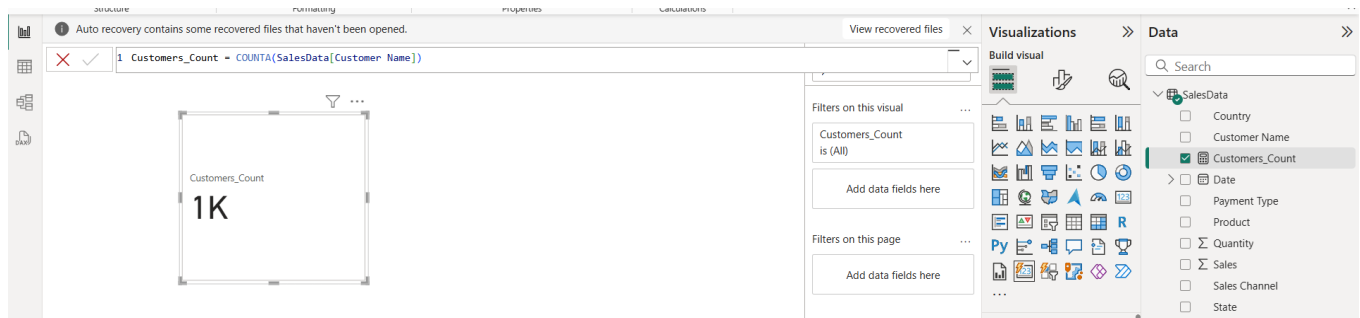
: - `Total_Transactions = COUNT(SalesData[Transaction ID])`



COUNTA : - Counts the number of non-empty rows (useful for text columns).

Example: - Counts the total number of unique customer names in the SalesData table, including blank values.

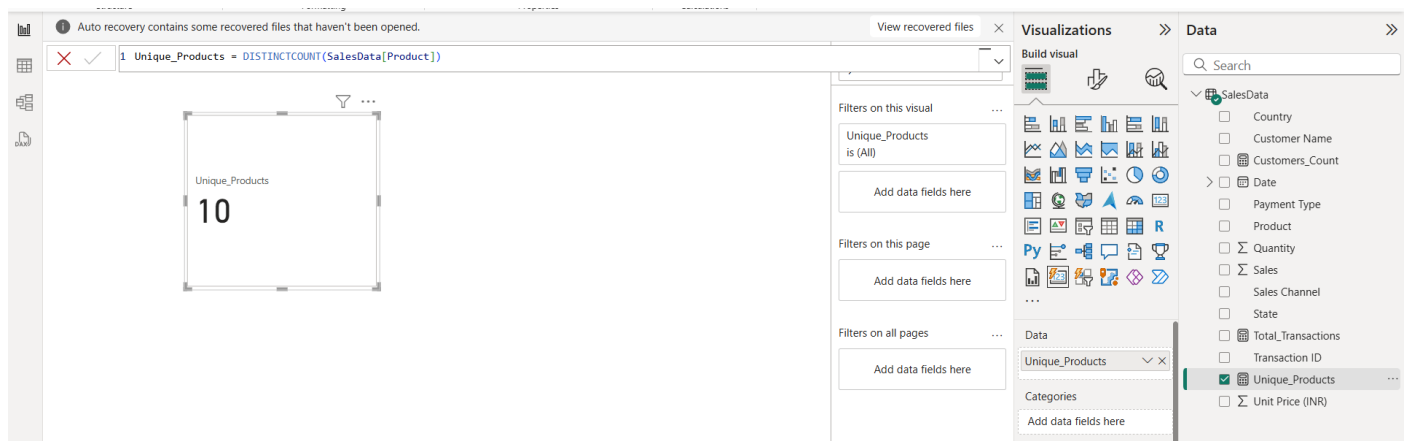
Customers_Count = COUNTA(SalesData[Customer Name])



DISTINCTCOUNT: - Counts the number of unique values in a column.

Example: - How many unique products are available in the dataset?

Unique_Products = DISTINCTCOUNT(SalesData[Product])



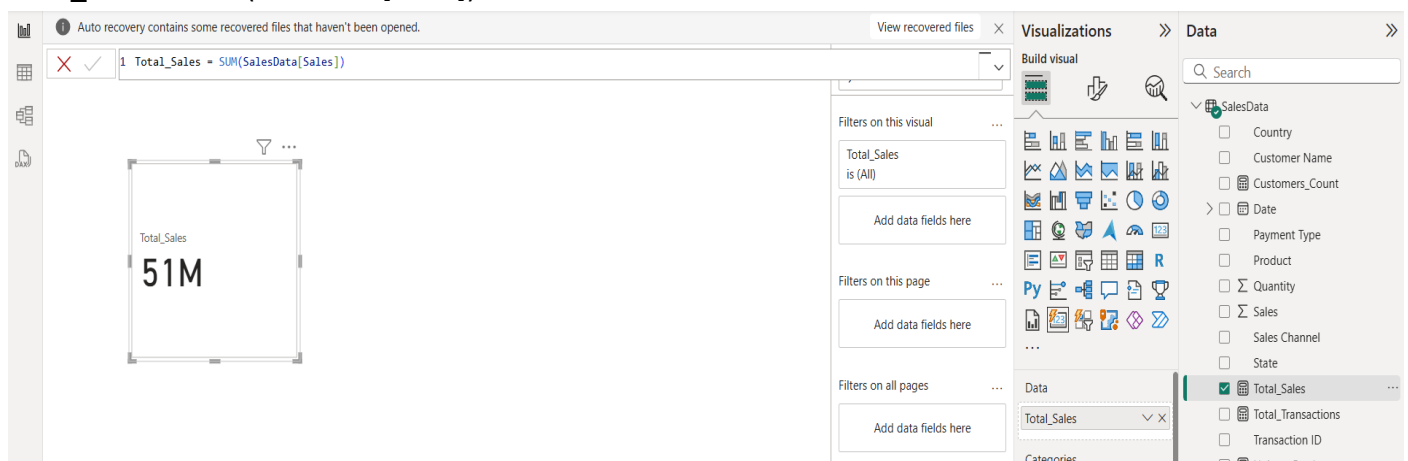
2) SUM Functions: -

These functions calculate the total sum of numeric columns.

SUM: Adds up all values in a column.

Example: - How to count the total number of transactions in the dataset?

Total_Sales = SUM(SalesData[Sales])

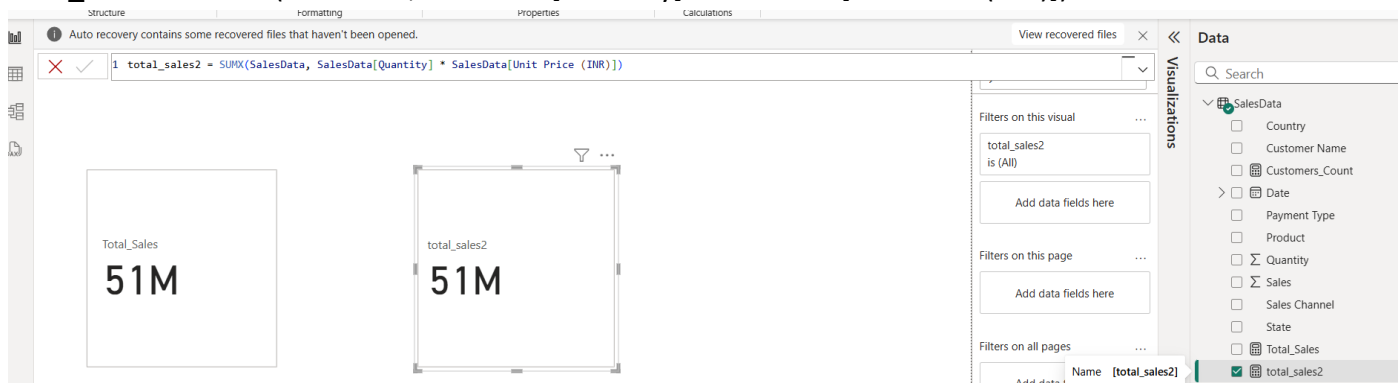


SUMX: Evaluates an expression for each row and then sums the result.

If you do not have a Sales column but have Quantity and Unit Price, you need SUMX:

Example: - How to calculate total sales dynamically when a separate sales column is not available

$\text{total_sales2} = \text{SUMX}(\text{SalesData}, \text{SalesData}[\text{Quantity}] * \text{SalesData}[\text{Unit Price (INR)}])$



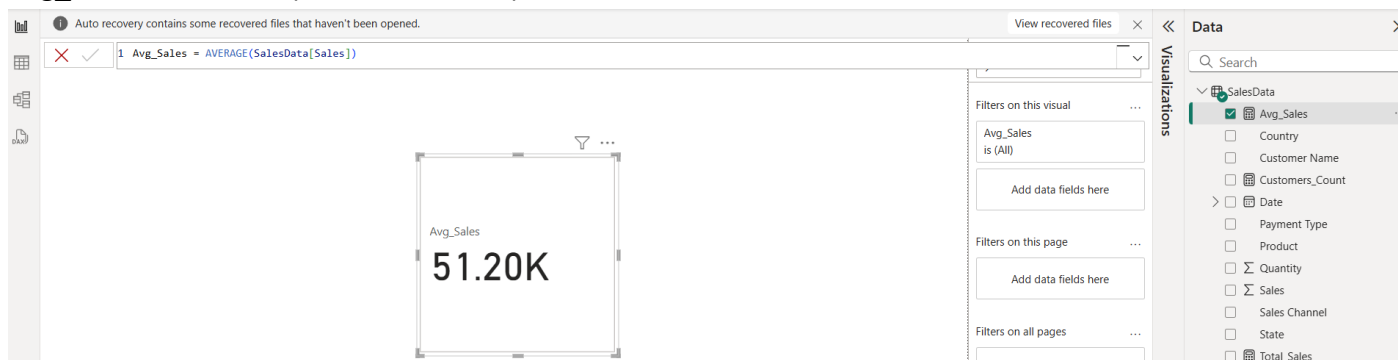
3) AVERAGE Functions: -

These functions calculate the average (mean) of a column's values.

AVERAGE: Returns the mean of a column.

Example: - How to calculate the average sales from the SalesData table?

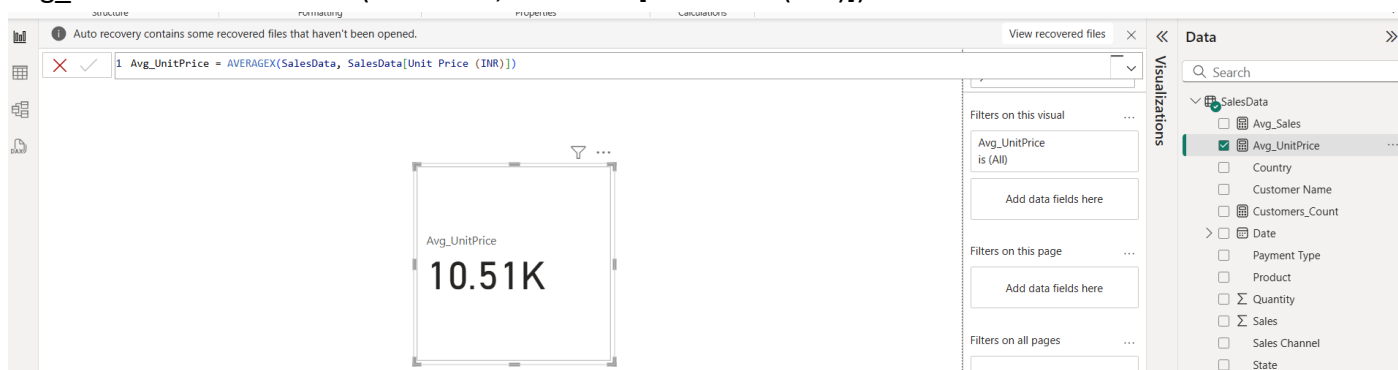
$\text{Avg_Sales} = \text{AVERAGE}(\text{SalesData}[\text{Sales}])$



AVERAGEX: Computes row-wise values and then averages them.

Example: - How to calculate the average unit price dynamically for each row in the SalesData table?

$\text{Avg_UnitPrice} = \text{AVERAGEX}(\text{SalesData}, \text{SalesData}[\text{Unit Price (INR)}])$



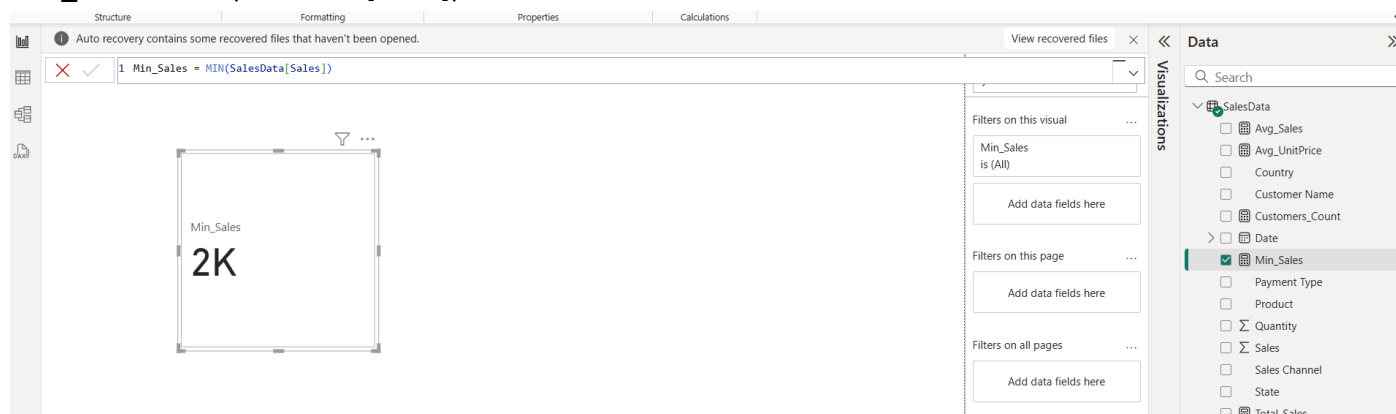
4) MIN and MAX Functions

These functions return the smallest and largest values in a column.

MIN: Finds the lowest value.

Example: - How to find the minimum sales value from the Sales column in the SalesData table?

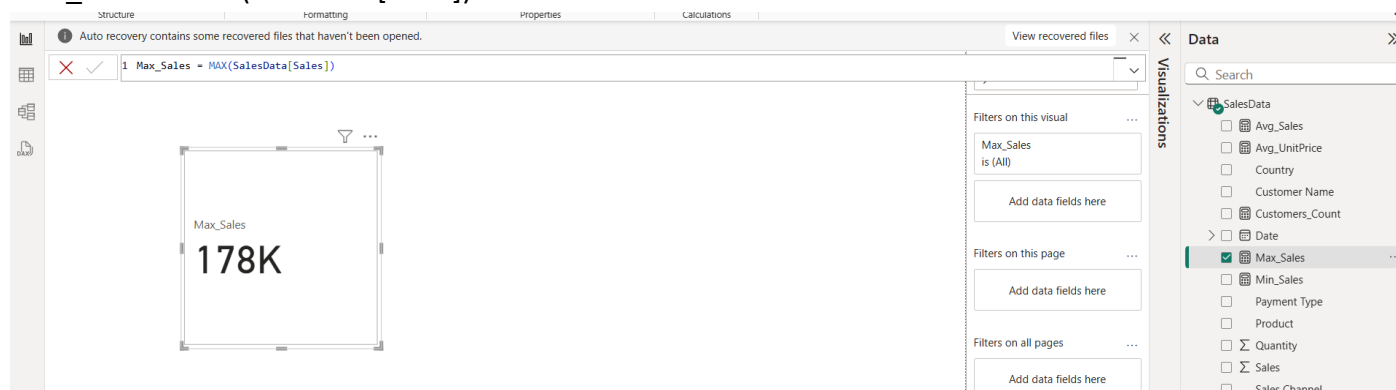
Min_Sales = MIN(SalesData[Sales])



MAX: Finds the highest value.

Example: - How to find the maximum sales value from the Sales column in the SalesData table?

Max_Sales = MAX(SalesData[Sales])

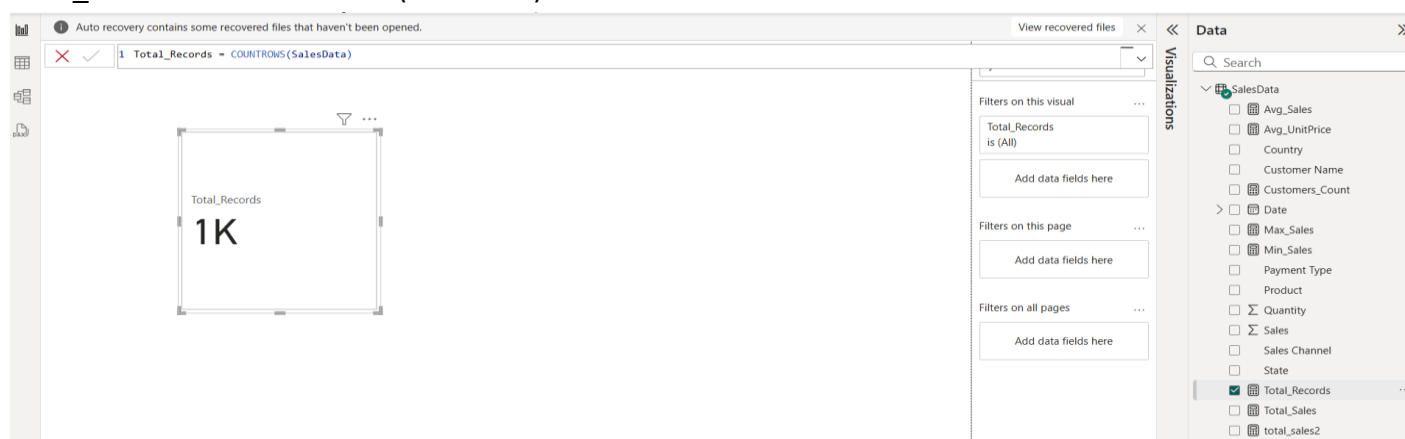


5) COUNTROWS

COUNTROWS: Counts the number of rows in a table.

Example: - How to count the total number of rows in the SalesData table?

Total_Records = COUNTROWS(SalesData)



Logical Functions

Logical functions in DAX are used to perform conditional checks, return boolean values (TRUE or FALSE), and create conditional calculations. These functions help in decision-making processes by evaluating expressions.

1) IF Function

The IF function evaluates a condition and returns one value if TRUE and another if FALSE.

Example: - If Sales is greater than 10,000, return "Yes", otherwise "No".

Discount_Applicable = IF(SalesData[Sales] > 10000, "Yes", "No")

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Data

1 Discount_Applicable = IF(SalesData[Sales] > 10000, "Yes", "No")

saction ID	Product	Quantity	Unit Price (INR)	Date	Customer Name	State	Country	Sales Channel	Payment Type	Sales	Discount_Applicable
20	Printer	6	2081	02 August 2024	Dharmajan Wable	Assam	India	Store	Cash	12486	Yes
27	External Hard Drive	8	5670	01 July 2024	Divyansh Ahluwalia	Mizoram	India	Store	Cash	45360	Yes
43	Tablet	3	10629	22 September 2024	Ryan Sidhu	Kerala	India	Store	Cash	31887	Yes
70	Smartphone	3	16998	30 December 2023	Inaaya Varughese	Kerala	India	Store	Cash	50994	Yes
71	Headphones	6	7330	25 June 2024	Divit Srivastava	Maharashtra	India	Store	Cash	43980	Yes
80	Desktop PC	2	7612	03 September 2024	Divij Seth	Mizoram	India	Store	Cash	15224	Yes
85	Printer	1	12305	22 June 2024	Nirvi Bhalla	Haryana	India	Store	Cash	12305	Yes
02	Smartphone	9	9197	03 August 2024	Jayan Balasubramanian	Punjab	India	Store	Cash	82773	Yes
14	Desktop PC	4	7074	11 April 2024	Prisha Chanda	Madhya Pradesh	India	Store	Cash	28296	Yes
35	Camera	2	4737	09 February 2024	Pihu Bakshi	Jharkhand	India	Store	Cash	9474	No
62	Camera	8	19800	03 April 2024	Vanya Chowdhury	Goa	India	Store	Cash	158400	Yes
66	External Hard Drive	2	17455	09 October 2023	Farhan Jhaveri	Assam	India	Store	Cash	34910	Yes
89	Tablet	6	11993	28 August 2024	Parinaaz Lall	Andhra Pradesh	India	Store	Cash	71958	Yes
90	Printer	1	13133	01 June 2024	Zain Lad	Odisha	India	Store	Cash	13133	Yes
98	Tablet	9	18074	12 June 2024	Anahi Wagle	Uttar Pradesh	India	Store	Cash	162666	Yes
07	External Hard Drive	4	3565	15 September 2024	Manjari Shenoy	Manipur	India	Store	Cash	14260	Yes
29	Printer	8	3175	05 November 2023	Purab Raja	Kerala	India	Store	Cash	25400	Yes
31	Smartphone	7	19589	08 December 2023	Suhana Konda	Sikkim	India	Store	Cash	137123	Yes
33	External Hard Drive	8	12350	01 November 2023	Ayesha Sarraf	Himachal Pradesh	India	Store	Cash	98800	Yes
44	Smartphone	8	12274	27 September 2023	Shray Jain	Chhattisgarh	India	Store	Cash	98192	Yes
52	Tablet	4	7666	08 July 2024	Fateh Aggarwal	Punjab	India	Store	Cash	30664	Yes
62	Camera	5	5715	05 February 2024	Ahana Jain	Jharkhand	India	Store	Cash	28575	Yes
76	Desktop PC	7	3588	12 May 2024	Alisha Arya	Andhra Pradesh	India	Store	Cash	25116	Yes
83	Laptop	1	18243	31 March 2024	Farhan Grover	Chhattisgarh	India	Store	Cash	18243	Yes
90	Printer	8	5704	08 November 2023	Dania Inchi	Andhra Pradesh	India	Store	Cash	45622	Yes

SalesData

Avg_Sales

Avg_UnitPrice

Country

Customer Name

Customers_Count

Date

Discount_Applicable

Max_Sales

Min_Sales

Payment Type

Product

Quantity

Sales

Sales Channel

State

Total_Records

Total_Sales

total_sales2

Total_Transactions

Transaction ID

Unique_Products

Nested IF

Example: -

If Sales > 20,000 → "Premium"

Else If Sales > 10,000 → "Standard"

Else → "Basic"

Sales_Category = IF(SalesData[Sales] > 20000, "Premium",

IF(SalesData[Sales] > 10000, "Standard", "Basic"))

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Data

1 Sales_Category = IF(SalesData[Sales] > 20000, "Premium",
2 IF(SalesData[Sales] > 10000, "Standard", "Basic"))

Product	Quantity	Unit Price (INR)	Date	Customer Name	State	Country	Sales Channel	Payment Type	Sales	Discount_Applicable	Sales_Category
Printer	6	2081	02 August 2024	Dharmajan Wable	Assam	India	Store	Cash	12486	Yes	Standard
External Hard Drive	8	5670	01 July 2024	Divyansh Ahluwalia	Mizoram	India	Store	Cash	45360	Yes	Premium
Tablet	3	10629	22 September 2024	Ryan Sidhu	Kerala	India	Store	Cash	31887	Yes	Premium
Smartphone	3	16998	30 December 2023	Inaaya Varughese	Kerala	India	Store	Cash	50994	Yes	Premium
Headphones	6	7330	25 June 2024	Divit Srivastava	Maharashtra	India	Store	Cash	43980	Yes	Premium
Desktop PC	2	7612	03 September 2024	Divij Seth	Mizoram	India	Store	Cash	15224	Yes	Standard
Printer	1	12305	22 June 2024	Nirvi Bhalla	Haryana	India	Store	Cash	12305	Yes	Standard
Smartphone	9	9197	03 August 2024	Jayan Balasubramanian	Punjab	India	Store	Cash	82773	Yes	Premium
Desktop PC	4	7074	11 April 2024	Prisha Chanda	Madhya Pradesh	India	Store	Cash	28296	Yes	Premium
Camera	2	4737	09 February 2024	Pihu Bakshi	Jharkhand	India	Store	Cash	9474	No	Basic
Camera	8	19800	03 April 2024	Vanya Chowdhury	Goa	India	Store	Cash	158400	Yes	Premium
External Hard Drive	2	17455	09 October 2023	Farhan Jhaveri	Assam	India	Store	Cash	34910	Yes	Premium
Tablet	6	11993	28 August 2024	Parinaaz Lall	Andhra Pradesh	India	Store	Cash	71958	Yes	Premium
Printer	1	13133	01 June 2024	Zain Lad	Odisha	India	Store	Cash	13133	Yes	Standard
Tablet	9	18074	12 June 2024	Anahi Wagle	Uttar Pradesh	India	Store	Cash	162666	Yes	Premium
External Hard Drive	4	3565	15 September 2024	Manjari Shenoy	Manipur	India	Store	Cash	14260	Yes	Standard
Printer	8	3175	05 November 2023	Purab Raja	Kerala	India	Store	Cash	25400	Yes	Premium
Smartphone	7	19589	08 December 2023	Suhana Konda	Sikkim	India	Store	Cash	137123	Yes	Premium
External Hard Drive	8	12350	01 November 2023	Ayesha Sarraf	Himachal Pradesh	India	Store	Cash	98800	Yes	Premium
Smartphone	8	12274	27 September 2023	Shray Jain	Chhattisgarh	India	Store	Cash	98192	Yes	Premium
Tablet	4	7666	08 July 2024	Fateh Aggarwal	Punjab	India	Store	Cash	30664	Yes	Premium
Camera	5	5715	05 February 2024	Ahana Jain	Jharkhand	India	Store	Cash	28575	Yes	Premium
Desktop PC	7	3588	12 May 2024	Alisha Arya	Andhra Pradesh	India	Store	Cash	25116	Yes	Premium

SalesData

Avg_Sales

Avg_UnitPrice

Country

Customer Name

Customers_Count

Date

Discount_Applicable

Max_Sales

Min_Sales

Payment Type

Product

Quantity

Sales

Sales Channel

State

Total_Records

Total_Sales

total_sales2

Total_Transactions

2) NOT Function

Example: -

If Sales is NOT greater than 10,000, return "Low Sales".

Not_High_Sales = IF(NOT(SalesData[Sales] > 10000), "Low Sales", "High Sales")

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1 Not_High_Sales = IF(NOT(SalesData[Sales] > 10000), "Low Sales", "High Sales")

Qty	Unit Price (INR)	Date	Customer Name	State	Country	Sales Channel	Payment Type	Sales	Discount Applicable	Sales Category	Not_High_Sales
6	2081	02 August 2024	Dharmajan Wable	Assam	India	Store	Cash	12486	Yes	Standard	High Sales
8	5670	01 July 2024	Divyansh Ahluwalia	Mizoram	India	Store	Cash	45360	Yes	Premium	High Sales
3	10629	22 September 2024	Ryan Sidhu	Kerala	India	Store	Cash	31887	Yes	Premium	High Sales
3	16998	30 December 2023	Inaaya Varughese	Kerala	India	Store	Cash	50994	Yes	Premium	High Sales
6	7330	25 June 2024	Divit Srivastava	Maharashtra	India	Store	Cash	43980	Yes	Premium	High Sales
2	7612	03 September 2024	Divij Seth	Mizoram	India	Store	Cash	15224	Yes	Standard	High Sales
1	12305	22 June 2024	Nirvi Bhalla	Haryana	India	Store	Cash	12305	Yes	Standard	High Sales
9	9197	03 August 2024	Jayan Balasubramanian	Punjab	India	Store	Cash	82773	Yes	Premium	High Sales
4	7074	11 April 2024	Prisha Chanda	Madhya Pradesh	India	Store	Cash	28296	Yes	Premium	High Sales
2	4737	09 February 2024	Pihu Bakshi	Jharkhand	India	Store	Cash	9474	No	Basic	Low Sales
8	19800	03 April 2024	Vanya Chowdhury	Goa	India	Store	Cash	158400	Yes	Premium	High Sales
2	17455	09 October 2023	Farhan Jhaveri	Assam	India	Store	Cash	34910	Yes	Premium	High Sales
6	11993	28 August 2024	Parinaaz Lall	Andhra Pradesh	India	Store	Cash	71958	Yes	Premium	High Sales
1	13133	01 June 2024	Zain Lad	Odisha	India	Store	Cash	13133	Yes	Standard	High Sales
9	18074	12 June 2024	Anahi Wagle	Uttar Pradesh	India	Store	Cash	162666	Yes	Premium	High Sales
4	3565	15 September 2024	Manjari Shenoy	Manipur	India	Store	Cash	14260	Yes	Standard	High Sales
8	3175	05 November 2023	Purab Raja	Kerala	India	Store	Cash	25400	Yes	Premium	High Sales
7	19589	08 December 2023	Suhana Konda	Sikkim	India	Store	Cash	137123	Yes	Premium	High Sales
8	12350	01 November 2023	Ayesha Sarraf	Himachal Pradesh	India	Store	Cash	98800	Yes	Premium	High Sales
8	12274	27 September 2023	Shray Jain	Chhattisgarh	India	Store	Cash	98192	Yes	Premium	High Sales
4	7666	08 July 2024	Fateh Aggarwal	Punjab	India	Store	Cash	30664	Yes	Premium	High Sales
5	5715	05 February 2024	Ahana Jain	Jharkhand	India	Store	Cash	28575	Yes	Premium	High Sales
7	3588	12 May 2024	Alisha Arya	Andhra Pradesh	India	Store	Cash	25116	Yes	Premium	High Sales
1	18243	31 March 2024	Farhan Grover	Chhattisgarh	India	Store	Cash	18243	Yes	Standard	High Sales

Search

SalesData

Avg_Sales

Avg_UnitPrice

Country

Customer Name

Customers_Count

Date

Discount_Applicable

Max_Sales

Min_Sales

Not_High_Sales

Payment Type

Product

Quantity

Sales

Sales Channel

Sales_Category

State

Total_Records

Total_Sales

total_sales2

Total Transactions

Text Functions

DAX provides several functions for manipulating text values, which can be useful for cleaning data, formatting strings, or extracting specific parts of text.

- Concatenation Functions
- Text Extraction Functions
- Case Conversion Functions
- Search & Replace Functions

Concatenation Functions

These functions combine multiple text values.

CONCATENATE Function: -

Joins two text values together.

Example: -full_name = CONCATENATE(SalesData[Product], " - " & SalesData[State])

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1 full_name = CONCATENATE(SalesData[Product], " - " & SalesData[State])

State	Country	Sales Channel	Payment Type	Sales	Discount Applicable	Sales Category	Not_High_Sales	full_name
Assam	India	Store	Cash	12486	Yes	Standard	High Sales	Printer - Assam
Assam	India	Store	Cash	45360	Yes	Premium	High Sales	External Hard Drive - Mizoram
Assam	India	Store	Cash	31887	Yes	Premium	High Sales	Tablet - Kerala
Assam	India	Store	Cash	50994	Yes	Premium	High Sales	Smartphone - Kerala
Assam	India	Store	Cash	43980	Yes	Premium	High Sales	Headphones - Maharashtra
Assam	India	Store	Cash	15224	Yes	Standard	High Sales	Desktop PC - Mizoram
Assam	India	Store	Cash	12305	Yes	Standard	High Sales	Printer - Haryana
Assam	India	Store	Cash	82773	Yes	Premium	High Sales	Smartphone - Punjab
Assam	India	Store	Cash	28296	Yes	Premium	High Sales	Desktop PC - Madhya Pradesh
Assam	India	Store	Cash	9474	No	Basic	Low Sales	Camera - Jharkhand
Assam	India	Store	Cash	158400	Yes	Premium	High Sales	Camera - Goa
Assam	India	Store	Cash	34910	Yes	Premium	High Sales	External Hard Drive - Assam
Assam	India	Store	Cash	71958	Yes	Premium	High Sales	Tablet - Andhra Pradesh
Assam	India	Store	Cash	13133	Yes	Standard	High Sales	Printer - Odisha
Assam	India	Store	Cash	162666	Yes	Premium	High Sales	Tablet - Uttar Pradesh
Assam	India	Store	Cash	14260	Yes	Standard	High Sales	External Hard Drive - Manipur
Assam	India	Store	Cash	25400	Yes	Premium	High Sales	Printer - Kerala
Assam	India	Store	Cash	137123	Yes	Premium	High Sales	Smartphone - Sikkim
Assam	India	Store	Cash	98800	Yes	Premium	High Sales	External Hard Drive - Himachal Pradesh
Assam	India	Store	Cash	98192	Yes	Premium	High Sales	Smartphone - Chhattisgarh
Assam	India	Store	Cash	30664	Yes	Premium	High Sales	Tablet - Punjab
Assam	India	Store	Cash	28575	Yes	Premium	High Sales	Camera - Jharkhand
Assam	India	Store	Cash	25116	Yes	Premium	High Sales	Desktop PC - Andhra Pradesh
Assam	India	Store	Cash	18243	Yes	Standard	High Sales	Laptop - Chhattisgarh
Assam	India	Store	Cash	45632	Yes	Premium	High Sales	Laptop - Andhra Pradesh

Using & Operator

Example: -

full_details = SalesData[Product] & " (" & SalesData[State] & ")"

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1 full_details = SalesData[Product] & " (" & SalesData[State] & ")"

State	Country	Sales Channel	Payment Type	Sales	Discount Applicable	Sales Category	Not_High_Sales	full_name	full_details
Assam	India	Store	Cash	12486	Yes	Standard	High Sales	Printer - Assam	Printer (Assam)
Assam	India	Store	Cash	45360	Yes	Premium	High Sales	External Hard Drive - Mizoram	External Hard Drive (Mizoram)
Kerala	India	Store	Cash	31887	Yes	Premium	High Sales	Tablet - Kerala	Tablet (Kerala)
Kerala	India	Store	Cash	50994	Yes	Premium	High Sales	Smartphone - Kerala	Smartphone (Kerala)
Maharashtra	India	Store	Cash	43980	Yes	Premium	High Sales	Headphones - Maharashtra	Headphones (Maharashtra)
Mizoram	India	Store	Cash	15224	Yes	Standard	High Sales	Desktop PC - Mizoram	Desktop PC (Mizoram)
Haryana	India	Store	Cash	12305	Yes	Standard	High Sales	Printer - Haryana	Printer (Haryana)
Punjab	India	Store	Cash	82773	Yes	Premium	High Sales	Smartphone - Punjab	Smartphone (Punjab)
Madhya Pradesh	India	Store	Cash	28296	Yes	Premium	High Sales	Desktop PC - Madhya Pradesh	Desktop PC (Madhya Pradesh)
Jharkhand	India	Store	Cash	9474	No	Basic	Low Sales	Camera - Jharkhand	Camera (Jharkhand)
Goa	India	Store	Cash	158400	Yes	Premium	High Sales	Camera - Goa	Camera (Goa)
Assam	India	Store	Cash	34910	Yes	Premium	High Sales	External Hard Drive - Assam	External Hard Drive (Assam)
Andhra Pradesh	India	Store	Cash	71958	Yes	Premium	High Sales	Tablet - Andhra Pradesh	Tablet (Andhra Pradesh)
Odisha	India	Store	Cash	13133	Yes	Standard	High Sales	Printer - Odisha	Printer (Odisha)
Uttar Pradesh	India	Store	Cash	162666	Yes	Premium	High Sales	Tablet - Uttar Pradesh	Tablet (Uttar Pradesh)
Manipur	India	Store	Cash	14260	Yes	Standard	High Sales	External Hard Drive - Manipur	External Hard Drive (Manipur)
Kerala	India	Store	Cash	25400	Yes	Premium	High Sales	Printer - Kerala	Printer (Kerala)
Sikkim	India	Store	Cash	137123	Yes	Premium	High Sales	Smartphone - Sikkim	Smartphone (Sikkim)
Himachal Pradesh	India	Store	Cash	98800	Yes	Premium	High Sales	External Hard Drive - Himachal Pradesh	External Hard Drive (Himachal Pradesh)
Chhattisgarh	India	Store	Cash	98192	Yes	Premium	High Sales	Smartphone - Chhattisgarh	Smartphone (Chhattisgarh)
Punjab	India	Store	Cash	30664	Yes	Premium	High Sales	Tablet - Punjab	Tablet (Punjab)
Jharkhand	India	Store	Cash	28575	Yes	Premium	High Sales	Camera - Jharkhand	Camera (Jharkhand)
Andhra Pradesh	India	Store	Cash	25116	Yes	Premium	High Sales	Desktop PC - Andhra Pradesh	Desktop PC (Andhra Pradesh)
Chhattisgarh	India	Store	Cash	18243	Yes	Standard	High Sales	Laptop - Chhattisgarh	Laptop (Chhattisgarh)
Andhra Pradesh	India	Store	Cash	45632	Yes	Premium	High Sales	Laptop - Andhra Pradesh	Laptop (Andhra Pradesh)

Case Conversion Functions

Used to convert text to uppercase or lowercase.

Convert to Uppercase

Example: -

uppercase_product = UPPER(SalesData[Product])

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1 uppercase_product = UPPER(SalesData[Product])

Sales Channel	Payment Type	Sales	Discount Applicable	Sales Category	Not_High_Sales	full_name	full_details	uppercase_product
Store	Cash	12486	Yes	Standard	High Sales	Printer - Assam	Printer (Assam)	PRINTER
Store	Cash	45360	Yes	Premium	High Sales	External Hard Drive - Mizoram	External Hard Drive (Mizoram)	EXTERNAL HARD DRIVE
Store	Cash	31887	Yes	Premium	High Sales	Tablet - Kerala	Tablet (Kerala)	TABLET
Store	Cash	50994	Yes	Premium	High Sales	Smartphone - Kerala	Smartphone (Kerala)	SMARTPHONE
Store	Cash	43980	Yes	Premium	High Sales	Headphones - Maharashtra	Headphones (Maharashtra)	HEADPHONES
Store	Cash	15224	Yes	Standard	High Sales	Desktop PC - Mizoram	Desktop PC (Mizoram)	DESKTOP PC
Store	Cash	12305	Yes	Standard	High Sales	Printer - Haryana	Printer (Haryana)	PRINTER
Store	Cash	82773	Yes	Premium	High Sales	Smartphone - Punjab	Smartphone (Punjab)	SMARTPHONE
Store	Cash	28296	Yes	Premium	High Sales	Desktop PC - Madhya Pradesh	Desktop PC (Madhya Pradesh)	DESKTOP PC
Store	Cash	9474	No	Basic	Low Sales	Camera - Jharkhand	Camera (Jharkhand)	CAMERA
Store	Cash	158400	Yes	Premium	High Sales	Camera - Goa	Camera (Goa)	CAMERA
Store	Cash	34910	Yes	Premium	High Sales	External Hard Drive - Assam	External Hard Drive (Assam)	EXTERNAL HARD DRIVE
Store	Cash	71958	Yes	Premium	High Sales	Tablet - Andhra Pradesh	Tablet (Andhra Pradesh)	TABLET
Store	Cash	13133	Yes	Standard	High Sales	Printer - Odisha	Printer (Odisha)	PRINTER
Store	Cash	162666	Yes	Premium	High Sales	Tablet - Uttar Pradesh	Tablet (Uttar Pradesh)	TABLET
Store	Cash	14260	Yes	Standard	High Sales	External Hard Drive - Manipur	External Hard Drive (Manipur)	EXTERNAL HARD DRIVE
Store	Cash	25400	Yes	Premium	High Sales	Printer - Kerala	Printer (Kerala)	PRINTER
Store	Cash	137123	Yes	Premium	High Sales	Smartphone - Sikkim	Smartphone (Sikkim)	SMARTPHONE
Store	Cash	98800	Yes	Premium	High Sales	External Hard Drive - Himachal Pradesh	External Hard Drive (Himachal Pradesh)	EXTERNAL HARD DRIVE
Store	Cash	98192	Yes	Premium	High Sales	Smartphone - Chhattisgarh	Smartphone (Chhattisgarh)	SMARTPHONE
Store	Cash	30664	Yes	Premium	High Sales	Tablet - Punjab	Tablet (Punjab)	TABLET
Store	Cash	28575	Yes	Premium	High Sales	Camera - Jharkhand	Camera (Jharkhand)	CAMERA
Store	Cash	25116	Yes	Premium	High Sales	Desktop PC - Andhra Pradesh	Desktop PC (Andhra Pradesh)	DESKTOP PC
Store	Cash	18243	Yes	Standard	High Sales	Laptop - Chhattisgarh	Laptop (Chhattisgarh)	LAPTOP
Store	Cash	45632	Yes	Premium	High Sales	Laptop - Andhra Pradesh	Laptop (Andhra Pradesh)	LAPTOP

Convert to Lowercase

Example: -

lowercase_product = LOWER(SalesData[Product])

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1 lowercase_product = LOWER(SalesData[Product])

Payment Type	Sales	Discount Applicable	Sales Category	Not High Sales	full_name	full_details	uppercase_product	lowercase_product
ish	12486	Yes	Standard	High Sales	Printer - Assam	Printer (Assam)	PRINTER	printer
ish	45360	Yes	Premium	High Sales	External Hard Drive - Mizoram	External Hard Drive (Mizoram)	EXTERNAL HARD DRIVE	external hard drive
ish	31887	Yes	Premium	High Sales	Tablet - Kerala	Tablet (Kerala)	TABLET	tablet
ish	50994	Yes	Premium	High Sales	Smartphone - Kerala	Smartphone (Kerala)	SMARTPHONE	smartphone
ish	43980	Yes	Premium	High Sales	Headphones - Maharashtra	Headphones (Maharashtra)	HEADPHONES	headphones
ish	15224	Yes	Standard	High Sales	Desktop PC - Mizoram	Desktop PC (Mizoram)	DESKTOP PC	desktop pc
ish	12305	Yes	Standard	High Sales	Printer - Haryana	Printer (Haryana)	PRINTER	printer
ish	82773	Yes	Premium	High Sales	Smartphone - Punjab	Smartphone (Punjab)	SMARTPHONE	smartphone
ish	28296	Yes	Premium	High Sales	Desktop PC - Madhya Pradesh	Desktop PC (Madhya Pradesh)	DESKTOP PC	desktop pc
ish	9474	No	Basic	Low Sales	Camera - Jharkhand	Camera (Jharkhand)	CAMERA	camera
ish	158400	Yes	Premium	High Sales	Camera - Goa	Camera (Goa)	CAMERA	camera
ish	34910	Yes	Premium	High Sales	External Hard Drive - Assam	External Hard Drive (Assam)	EXTERNAL HARD DRIVE	external hard drive
ish	71958	Yes	Premium	High Sales	Tablet - Andhra Pradesh	Tablet (Andhra Pradesh)	TABLET	tablet
ish	13133	Yes	Standard	High Sales	Printer - Odisha	Printer (Odisha)	PRINTER	printer
ish	162666	Yes	Premium	High Sales	Tablet - Uttar Pradesh	Tablet (Uttar Pradesh)	TABLET	tablet
ish	14260	Yes	Standard	High Sales	External Hard Drive - Manipur	External Hard Drive (Manipur)	EXTERNAL HARD DRIVE	external hard drive
ish	25400	Yes	Premium	High Sales	Printer - Kerala	Printer (Kerala)	PRINTER	printer
ish	137123	Yes	Premium	High Sales	Smartphone - Sikkim	Smartphone (Sikkim)	SMARTPHONE	smartphone
ish	98800	Yes	Premium	High Sales	External Hard Drive - Himachal Pradesh	External Hard Drive (Himachal Pradesh)	EXTERNAL HARD DRIVE	external hard drive
ish	98192	Yes	Premium	High Sales	Smartphone - Chhattisgarh	Smartphone (Chhattisgarh)	SMARTPHONE	smartphone
ish	30664	Yes	Premium	High Sales	Tablet - Punjab	Tablet (Punjab)	TABLET	tablet
ish	28575	Yes	Premium	High Sales	Camera - Jharkhand	Camera (Jharkhand)	CAMERA	camera
ish	25116	Yes	Premium	High Sales	Desktop PC - Andhra Pradesh	Desktop PC (Andhra Pradesh)	DESKTOP PC	desktop pc
ish	18243	Yes	Standard	High Sales	Laptop - Chhattisgarh	Laptop (Chhattisgarh)	LAPTOP	laptop
ish	45632	Yes	Premium	High Sales	Laptop - Andhra Pradesh	Laptop (Andhra Pradesh)	LAPTOP	laptop

Text Extraction Functions

These functions extract part of a text string.

Extract First 3 Letters (LEFT)

Example: -

short_state = LEFT(SalesData[State], 3)

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 short_state = LEFT(SalesData[State], 3)

Sales	Discount Applicable	Sales Category	Not High Sales	full_name	full_details	uppercase_product	lowercase_product	short_state
12486	Yes	Standard	High Sales	Printer - Assam	Printer (Assam)	PRINTER	printer	Ass
45360	Yes	Premium	High Sales	External Hard Drive - Mizoram	External Hard Drive (Mizoram)	EXTERNAL HARD DRIVE	external hard drive	Miz
31887	Yes	Premium	High Sales	Tablet - Kerala	Tablet (Kerala)	TABLET	tablet	Ker
50994	Yes	Premium	High Sales	Smartphone - Kerala	Smartphone (Kerala)	SMARTPHONE	smartphone	Ker
43980	Yes	Premium	High Sales	Headphones - Maharashtra	Headphones (Maharashtra)	HEADPHONES	headphones	Mah
15224	Yes	Standard	High Sales	Desktop PC - Mizoram	Desktop PC (Mizoram)	DESKTOP PC	desktop pc	Miz
12305	Yes	Standard	High Sales	Printer - Haryana	Printer (Haryana)	PRINTER	printer	Har
82773	Yes	Premium	High Sales	Smartphone - Punjab	Smartphone (Punjab)	SMARTPHONE	smartphone	Pun
28296	Yes	Premium	High Sales	Desktop PC - Madhya Pradesh	Desktop PC (Madhya Pradesh)	DESKTOP PC	desktop pc	Mad
9474	No	Basic	Low Sales	Camera - Jharkhand	Camera (Jharkhand)	CAMERA	camera	Jha
158400	Yes	Premium	High Sales	Camera - Goa	Camera (Goa)	CAMERA	camera	Goa
34910	Yes	Premium	High Sales	External Hard Drive - Assam	External Hard Drive (Assam)	EXTERNAL HARD DRIVE	external hard drive	Ass
71958	Yes	Premium	High Sales	Tablet - Andhra Pradesh	Tablet (Andhra Pradesh)	TABLET	tablet	And
13133	Yes	Standard	High Sales	Printer - Odisha	Printer (Odisha)	PRINTER	printer	Odi
162666	Yes	Premium	High Sales	Tablet - Uttar Pradesh	Tablet (Uttar Pradesh)	TABLET	tablet	Utt
14260	Yes	Standard	High Sales	External Hard Drive - Manipur	External Hard Drive (Manipur)	EXTERNAL HARD DRIVE	external hard drive	Man
25400	Yes	Premium	High Sales	Printer - Kerala	Printer (Kerala)	PRINTER	printer	Ker
137123	Yes	Premium	High Sales	Smartphone - Sikkim	Smartphone (Sikkim)	SMARTPHONE	smartphone	Sik
98800	Yes	Premium	High Sales	External Hard Drive - Himachal Pradesh	External Hard Drive (Himachal Pradesh)	EXTERNAL HARD DRIVE	external hard drive	Him
98192	Yes	Premium	High Sales	Smartphone - Chhattisgarh	Smartphone (Chhattisgarh)	SMARTPHONE	smartphone	Chh
30664	Yes	Premium	High Sales	Tablet - Punjab	Tablet (Punjab)	TABLET	tablet	Pun
28575	Yes	Premium	High Sales	Camera - Jharkhand	Camera (Jharkhand)	CAMERA	camera	Jha
25116	Yes	Premium	High Sales	Desktop PC - Andhra Pradesh	Desktop PC (Andhra Pradesh)	DESKTOP PC	desktop pc	And
18243	Yes	Standard	High Sales	Laptop - Chhattisgarh	Laptop (Chhattisgarh)	LAPTOP	laptop	Chh
45632	Yes	Premium	High Sales	Laptop - Andhra Pradesh	Laptop (Andhra Pradesh)	LAPTOP	laptop	And

Extract Last 4 Letters (RIGHT)

Example: -

last_4_chars = RIGHT(SalesData[State], 4)

Auto recovery contains some recovered files that haven't been opened.

1 last_4_chars = RIGHT(SalesData[State], 4)

unt_Applicable	Sales_Category	Not_High_Sales	full_name	full_details	uppercase_product	lowercase_product	short_state	last_4_chars
	Standard	High Sales	Printer - Assam	Printer (Assam)	PRINTER	printer	Ass	ssam
	Premium	High Sales	External Hard Drive - Mizoram	External Hard Drive (Mizoram)	EXTERNAL HARD DRIVE	external hard drive	Miz	oram
	Premium	High Sales	Tablet - Kerala	Tablet (Kerala)	TABLET	tablet	Ker	rala
	Premium	High Sales	Smartphone - Kerala	Smartphone (Kerala)	SMARTPHONE	smartphone	Ker	rala
	Premium	High Sales	Headphones - Maharashtra	Headphones (Maharashtra)	HEADPHONES	headphones	Mah	htra
	Standard	High Sales	Desktop PC - Mizoram	Desktop PC (Mizoram)	DESKTOP PC	desktop pc	Miz	oram
	Standard	High Sales	Printer - Haryana	Printer (Haryana)	PRINTER	printer	Har	yana
	Premium	High Sales	Smartphone - Punjab	Smartphone (Punjab)	SMARTPHONE	smartphone	Pun	njab
	Premium	High Sales	Desktop PC - Madhya Pradesh	Desktop PC (Madhya Pradesh)	DESKTOP PC	desktop pc	Mad	desh
	Basic	Low Sales	Camera - Jharkhand	Camera (Jharkhand)	CAMERA	camera	Jha	hand
	Premium	High Sales	Camera - Goa	Camera (Goa)	CAMERA	camera	Goa	Goa
	Premium	High Sales	External Hard Drive - Assam	External Hard Drive (Assam)	EXTERNAL HARD DRIVE	external hard drive	Ass	ssam
	Premium	High Sales	Tablet - Andhra Pradesh	Tablet (Andhra Pradesh)	TABLET	tablet	And	desh
	Standard	High Sales	Printer - Odisha	Printer (Odisha)	PRINTER	printer	Odi	isha
	Premium	High Sales	Tablet - Uttar Pradesh	Tablet (Uttar Pradesh)	TABLET	tablet	Utt	desh
	Standard	High Sales	External Hard Drive - Manipur	External Hard Drive (Manipur)	EXTERNAL HARD DRIVE	external hard drive	Man	ipur

Extract Middle Part (MID)

Example: -

middle_chars = MID(SalesData[State], 2, 4)

Auto recovery contains some recovered files that haven't been opened.

1 middle_chars = MID(SalesData[State], 2, 4)

Sales_Category	Not_High_Sales	full_name	full_details	uppercase_product	lowercase_product	short_state	last_4_chars	middle_chars
Standard	High Sales	Printer - Assam	Printer (Assam)	PRINTER	printer	Ass	ssam	ssam
Premium	High Sales	External Hard Drive - Mizoram	External Hard Drive (Mizoram)	EXTERNAL HARD DRIVE	external hard drive	Miz	oram	izor
Premium	High Sales	Tablet - Kerala	Tablet (Kerala)	TABLET	tablet	Ker	rala	eral
Premium	High Sales	Smartphone - Kerala	Smartphone (Kerala)	SMARTPHONE	smartphone	Ker	rala	eral
Premium	High Sales	Headphones - Maharashtra	Headphones (Maharashtra)	HEADPHONES	headphones	Mah	htra	ahar
Standard	High Sales	Desktop PC - Mizoram	Desktop PC (Mizoram)	DESKTOP PC	desktop pc	Miz	oram	izor
Standard	High Sales	Printer - Haryana	Printer (Haryana)	PRINTER	printer	Har	yana	arya
Premium	High Sales	Smartphone - Punjab	Smartphone (Punjab)	SMARTPHONE	smartphone	Pun	njab	unja
Premium	High Sales	Desktop PC - Madhya Pradesh	Desktop PC (Madhya Pradesh)	DESKTOP PC	desktop pc	Mad	desh	adhy
Basic	Low Sales	Camera - Jharkhand	Camera (Jharkhand)	CAMERA	camera	Jha	hand	hark
Premium	High Sales	Camera - Goa	Camera (Goa)	CAMERA	camera	Goa	Goa	oa
Premium	High Sales	External Hard Drive - Assam	External Hard Drive (Assam)	EXTERNAL HARD DRIVE	external hard drive	Ass	ssam	ssam
Premium	High Sales	Tablet - Andhra Pradesh	Tablet (Andhra Pradesh)	TABLET	tablet	And	desh	ndhr
Standard	High Sales	Printer - Odisha	Printer (Odisha)	PRINTER	printer	Odi	isha	dish
Premium	High Sales	Tablet - Uttar Pradesh	Tablet (Uttar Pradesh)	TABLET	tablet	Utt	desh	ttar
Standard	High Sales	External Hard Drive - Manipur	External Hard Drive (Manipur)	EXTERNAL HARD DRIVE	external hard drive	Man	ipur	anip

Search & Replace Functions

SEARCH Function

Finds the position of a substring in a text.

Syntax:

SEARCH(search_text, within_text, [start_position], [not_found_value])

Example: -

position = SEARCH("Laptop", SalesData[Product], 1, -1)

Returns the position where "Laptop" starts.

Returns -1 if "Laptop" is not found.

Auto recovery contains some recovered files that haven't been opened.

1 position = SEARCH("Laptop", SalesData[Product], 1, -1)

ory	Not_High_Sales	full_name	full_details	uppercase_product	lowercase_product	short_state	last_4_chars	middle_chars	position
	High Sales	Printer - Assam	Printer (Assam)	PRINTER	printer	Ass	ssam	ssam	-1
	High Sales	External Hard Drive - Mizoram	External Hard Drive (Mizoram)	EXTERNAL HARD DRIVE	external hard drive	Miz	oram	izor	-1
	High Sales	Tablet - Kerala	Tablet (Kerala)	TABLET	tablet	Ker	rala	eral	-1
	High Sales	Smartphone - Kerala	Smartphone (Kerala)	SMARTPHONE	smartphone	Ker	rala	eral	-1
	High Sales	Headphones - Maharashtra	Headphones (Maharashtra)	HEADPHONES	headphones	Mah	htra	ahar	-1
	High Sales	Desktop PC - Mizoram	Desktop PC (Mizoram)	DESKTOP PC	desktop pc	Miz	oram	izor	-1
	High Sales	Printer - Haryana	Printer (Haryana)	PRINTER	printer	Har	yana	arya	-1
	High Sales	Smartphone - Punjab	Smartphone (Punjab)	SMARTPHONE	smartphone	Pun	njab	unja	-1
	High Sales	Desktop PC - Madhya Pradesh	Desktop PC (Madhya Pradesh)	DESKTOP PC	desktop pc	Mad	desh	adhy	-1
	Low Sales	Camera - Jharkhand	Camera (Jharkhand)	CAMERA	camera	Jha	hand	hark	-1
	High Sales	Camera - Goa	Camera (Goa)	CAMERA	camera	Goa	Goa	oa	-1